Unit II – Mathematical models for decision making

Structure of mathematical models

What Is a Mathematical Model?

- **Definition**
- A **mathematical model** is a **selective abstraction** of a real system:
- It simplifies reality to focus only on elements **relevant to the investigation**.
- It helps analyze and understand the **operating behavior** of complex systems.
- *Einstein's wisdom*: "Everything should be made as simple as possible, but not simpler."

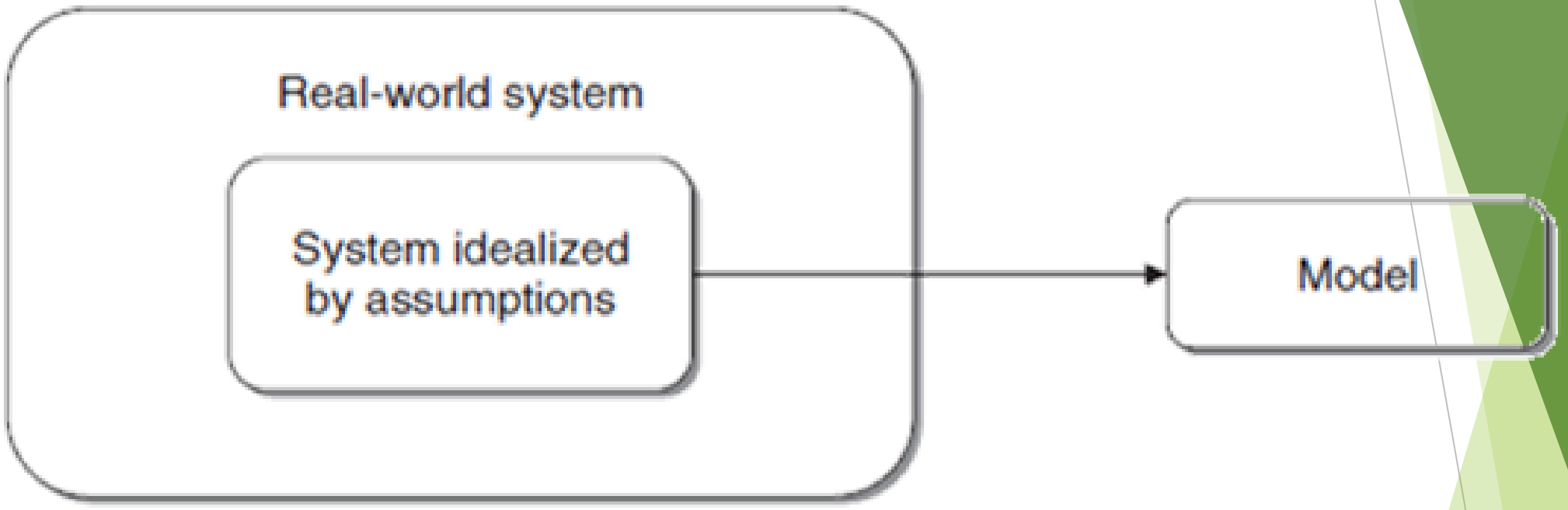
- **Shared Features of Mathematical Models**
- Despite differences across domains (physics, architecture, economics), most models share these traits:

Feature	Description
Abstraction	Focuses on key elements, ignoring irrelevant details
Purpose-driven	Tailored to answer specific questions or support decisions
Simplification	Reduces complexity while preserving essential behavior
Predictive capability	Allows simulation of outcomes under different scenarios
Modifiability	Can be adjusted to test hypotheses or reflect changes

- **Conceptual Analogy: Thought Experiments**
- **Gendanken experiment** (thought experiment):
  - Popular in physics (e.g., Einstein's relativity).
  - Involves mentally simulating a phenomenon and testing consequences of hypothetical changes.
- **Business Analogy**:
  - *What-if analysis* in spreadsheets:
    - E.g., How does changing invoice payment terms (90 vs. 60 days) affect cash flow?
- Example: Budget Model in a Spreadsheet

Scenario	Payment Terms	Impact on Cash Flow
Base Case	90 days	Delayed inflow
Modified Case	60 days	Faster inflow

- This simple model helps decision-makers **simulate outcomes** and **adjust strategies**.



A model is a selective abstraction of reality

- According to their characteristics, models can be divided into *iconic*, *analogical* and *symbolic*
- **Types of Models: Iconic, Analogical, Symbolic**
- **1. Iconic Models**
  - **Nature**: Physical, material representation.
  - **Purpose**: Replicates appearance and structure of the real system.
  - **Example**: A scale model of a city neighborhood.
  - **Use Case**: Urban planning, architecture, product design.
- **2. Analogical Models**
  - **Nature**: Physical representation using analogy.
  - **Purpose**: Simulates behavior through analogous systems.
  - **Example**: A wind tunnel simulating vehicle aerodynamics.
  - **Use Case**: Engineering, physics, environmental modeling.

- **3. Symbolic Models**
  - **Nature**: Abstract, mathematical or logical representation.
  - **Purpose**: Describes system behavior using variables, equations, algorithms.
  - **Example**: A budget model using formulas in a spreadsheet.
  - **Use Case**: Economics, decision support systems, simulations.

Comparison Table

Model Type	Representation	Behavior Simulation	Example	Domain Use
Iconic	Physical copy	Direct imitation	Scale model of a city	Architecture, design
Analogical	Physical analogy	Indirect simulation	Wind tunnel for car aerodynamics	Engineering, physics
Symbolic	Abstract	Mathematical modeling	Spreadsheet budget model	Economics, DSS, science

Types of Mathematical Models

Type	Structure	Example
Deterministic	Fixed rules, no randomness	Newton's laws of motion
Stochastic	Includes randomness/probability	Stock market models
Static	Time-independent	Budget allocation model
Dynamic	Evolves over time	Population growth equations
Discrete	Step-based, countable states	Queueing systems
Continuous	Smooth, unbroken variables	Fluid dynamics

- A further relevant distinction concerns the probabilistic nature of models, which can be either *stochastic* or *deterministic*.
- **Stochastic vs. Deterministic Models**
- **1. Stochastic Models**
  - **Definition**: Include input variables that are **random** and follow **probability distributions**.
  - **Behavior**: Outcomes vary due to uncertainty in inputs.
- **Examples**:
  - **Predictive models** (e.g., forecasting demand or risk).
  - **Queueing models** (e.g., waiting lines in service systems).
- **Use Case**: When randomness is central to system behavior.

- **2. Deterministic Models**
  - **Definition**: Assume all input data are **known with certainty**.
  - **Behavior**: Produces a single, predictable outcome for given inputs.
- **Examples**:
  - Optimization models (e.g., linear programming).
  - Budget planning with fixed parameters.
- **Use Case**: When randomness is negligible or manageable.
- *Note*: Even deterministic models can incorporate uncertainty using:
  - **Sensitivity analysis**: Tests how changes in inputs affect outcomes.
  - **Scenario analysis**: Explores multiple future states.
  - **What-if analysis**: Simulates hypothetical changes.
- **Key Takeaway**
- **Stochastic models** embrace uncertainty and simulate variability.
- **Deterministic models** simplify complexity by assuming certainty — but can still test robustness through analytical techniques.

Feature	Stochastic Model	Deterministic Model
Input Data	Random variables with probability	Known and fixed values
Output	Probabilistic, multiple possible outcomes	Single predictable outcome
Uncertainty Handling	Built-in via distributions	External via sensitivity/scenario analysis
Examples	Predictive analytics, queueing systems	Scheduling, budgeting, optimization
Use Case	Systems with inherent randomness	Systems with stable or controllable inputs

- A further distinction concerns the temporal dimension in a mathematical model, which can be either *static* or *dynamic*.
- **Temporal Dimension in Mathematical Models**
- **1. Static Models**
- **Definition**: Analyze a system at a **single point in time**.
- **Characteristics**:
  - No evolution or time-based changes.
  - Snapshot-style analysis.
- **Example**:
  - A one-time cost-benefit analysis.
  - A linear programming model for a single production cycle.
- **2. Dynamic Models**
- **Definition**: Analyze a system **over time**, across multiple stages.
- **Characteristics**:
  - Capture changes, feedback, and evolution.
  - Support sequential decision-making.

- Types of Dynamic Models:

Type	Description
Discrete-Time Models	Time is divided into fixed intervals (e.g., days, months). Status is observed at the start or end of each interval. Common in business intelligence.
Continuous-Time Models	Time flows continuously. System status can change at any moment. Common in physics, engineering, and control systems.

- Comparison Table

Feature	Static Model	Dynamic Model
Time Consideration	Single time point	Multiple time stages
System Evolution	Not modeled	Modeled over time
Decision Scope	One-time decision	Sequential decisions
Examples	Cost analysis, snapshot optimization	Inventory planning, forecasting
Subtypes	N/A	Discrete-time, Continuous-time

- **Key Takeaway**
- **Static models** offer simplicity and clarity for one-time decisions.
- **Dynamic models** provide depth and realism for evolving systems — especially in business intelligence, forecasting, and operations.

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Development of a model

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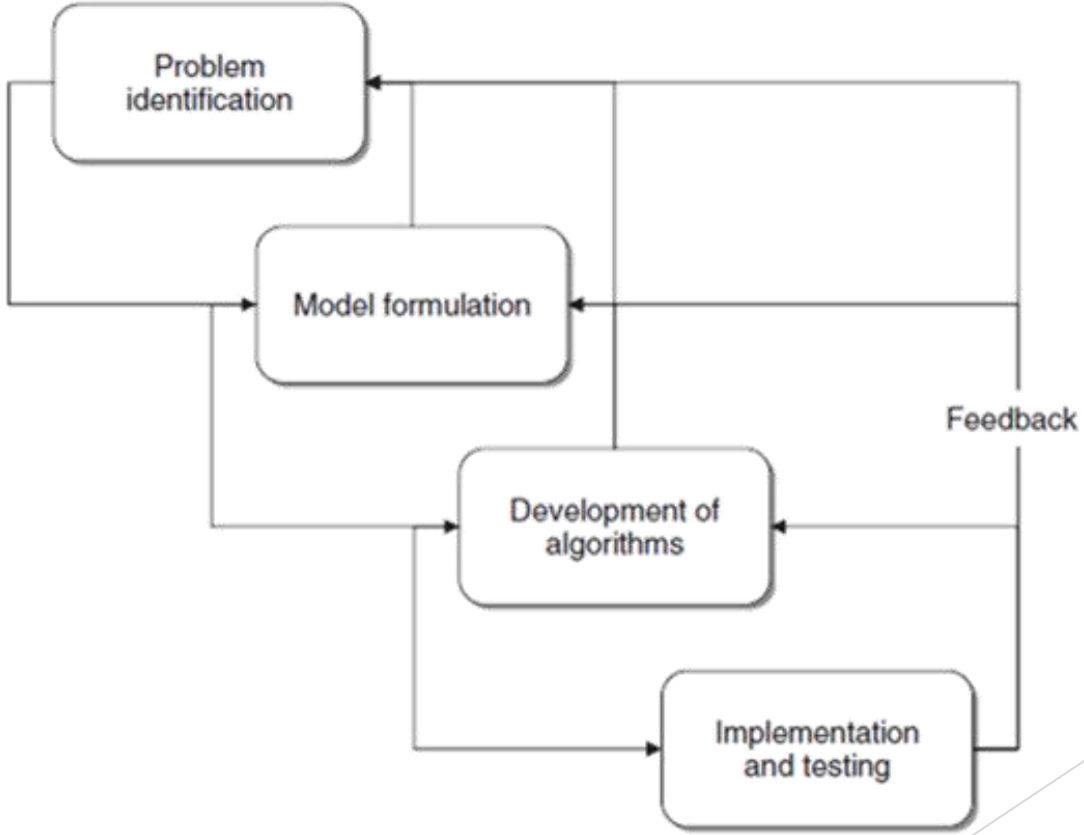
- **Development of a model**
- It is possible to break down the development of a mathematical model for decision making into four primary phases.
- The figure also includes a *feedback* mechanism which takes into account the possibility of changes and revisions of the model.



- *Phases in the development of mathematical models for decision making*

- **Problem Identification (Simplified for Students)**
- **Step 1: Spot the problem clearly** Before solving, you must know exactly what the problem is.
- **Step 2: Look at the symptoms** - Symptoms are signs that something is wrong. Example:
  - If a company has **too much stock** and turnover is high, that's a warning sign.
- **Step 3: Ask "Why?"** Find the root cause behind the symptom.
  - Managers' opinions can help explain what's happening.
- **Step 4: Example cause** In this case, the extra stock might be due to a **poor production plan**.
- In short: **Identify the problem** → **Study the symptoms** → **Ask why** → **Find the cause.**

- **Model Formulation (Simplified for Students)**
- Once you know the problem, the next step is to **build a model** (a simplified mathematical representation of the system). This helps in analyzing and making decisions.
- **Key Elements of a Model**
- **Time Horizon (When?)**
  - Models usually include time.
  - Example:
    - Tactical plan → production rate each week for a year.
    - Operational schedule → tasks assigned daily to each production line.
  - So, the time span depends on the type of problem.

- **Evaluation Criteria (How do we judge success?)**
  - Define measurable indicators to compare decisions.
  - Examples:
    - 💰 Costs and profits
    - ⚡ Efficiency and service level
    - ✅ Quality of products/services
    - 🔄 Flexibility of operations
    - 🎯 Reliability in meeting goals
- **Decision Variables (What choices can we make?)**
  - These represent possible decisions.
  - Example: In production planning → how much of each product to make, in each process, during each time period.

- **Numerical Parameters (What numbers do we need?)**
  - These are fixed values the model needs.
  - Example:
    - Available capacity for each process
    - How much capacity each product consumes
- **Mathematical Relationships (How do things connect?)**
  - Show how variables, parameters, and performance indicators are linked.
  - Can be:
    - **Deterministic** (fixed, predictable)
    - **Probabilistic** (involving chance/uncertainty)
  - Balance between **accuracy** and **simplicity**:
    - A simpler model may ignore small details but is easier to solve and adapt.
- **In Short:**
- **Identify the problem → Build a model → Include time, criteria, decisions, numbers, and relationships → Balance accuracy with simplicity.**

- **Development of Algorithms (Simplified for Students)**
- After building a **mathematical model**, the next step is to **solve it**.
- To do this, we need an **algorithm** (a step-by-step method) that can find the best decision or solution.
- Usually, this algorithm is implemented using a **software tool** (either developed or purchased).
- The analyst (the person working on the model) must know different solution methods and understand their strengths and limitations.
- **In Short:**
- **Model → Algorithm → Software → Best Decision**

- **Implementation and Test (Simplified for Students)**
- After the model is built, it must be **put into action (implemented)** and **tested**.
- First, check that the **data and numbers** used in the model are correct. These usually come from databases (like a data warehouse).
- Run the model to get the first results.
- Then, **validate** the model by asking experts and decision makers if the results make sense.
- **Things to Check During Testing**
- **Plausibility** – Do the results look realistic?
- **Consistency** – Are the results still logical even when extreme values are used?
- **Stability** – Do small changes in input data lead to stable results (not wild swings)?
- **In Short:**
- **Build → Implement → Test → Validate → Check realism, consistency, and stability.**

Unit II – Mathematical models for decision making

Classes of models

- **Classes of Mathematical Models for Decision Making**
- Different problems need different types of models. Each model helps in making decisions in its own way.
- **Main Types of Models**
- **Predictive Models**
- **Pattern Recognition & Learning Models**
- **Optimization Models**
- **Project Management Models**
- **Risk Analysis Models**
- **Waiting Line (Queue) Models**
- **In Short:**
- **Different models = different decision problems → choose the right model for the right situation.**

- **Predictive Models**
- Many business models need **future data** (like demand, prices, or new technologies).
- **Predictive models** help estimate or forecast these future events.
- They are important because predictions are used in almost every part of a company:
  - Strategy, Research & Development, Administration & Control, Marketing, Production, Logistics
- **Types of Predictive Models**
- **Explanatory Models**
  - Show the relationship between one main outcome (**dependent variable**) and other factors (**independent variables**).
  - Examples:
    - **Regression models** → predict sales based on advertising spend.
    - **Classification models** → group data (like “spam” vs “not spam”).

- **Time Series Models**
  - Look at data over time to find patterns.
  - Example: Predicting electricity demand by studying usage patterns across days or months.
- **In Short:**
- **Predictive models = tools to forecast the future → used everywhere in business → two main types: explanatory & time series.**

- **Pattern Recognition & Machine Learning Models (Simplified)**
- **What they do**
- Goal: Help machines **learn from past data** and use that knowledge for the future.
- Just like humans learn from experience, machines can also recognize patterns and make decisions.
- **Where they are used**
- Image, sound, and text recognition (e.g., face unlock, speech-to-text).
- Medical diagnosis (e.g., detecting diseases).
- Marketing (e.g., grouping customers by behavior).
- Manufacturing (e.g., controlling processes).
- Fraud detection and anomaly spotting.

- **Main Objectives**
- **Interpretation Models**
  - Find **patterns** in data and explain them with simple rules.
  - Example: Rules that separate “spam” emails from “non-spam.”
- **Prediction Models**
  - Use data to **predict future values**.
  - Example: Predicting house prices based on size, location, etc.
- **Types of Learning**
- **Supervised Learning**
  - Data has a **target label** (like “spam” or “not spam”).
  - Examples: Classification models, regression models.

- **Unsupervised Learning**
  - Data has **no labels**. The goal is to find patterns or groups.
  - Examples:
    - **Clustering** → grouping similar records together.
    - **Association rules** → finding relationships (e.g., “people who buy bread often buy butter”).
- **In Short:**
- **Machines learn from data → recognize patterns → predict outcomes → supervised (with labels) or unsupervised (without labels).**

- **Optimization Models (Simplified)**
- **What they are**
- Companies often face problems where they must **choose the best option** among many.
- The goal is to **minimize costs** or **maximize benefits**.
- This process is called **optimization**.
- **Why they matter**
- Resources (like people, money, materials, machines) are always **limited**.
- Optimization models help decide how to use these resources in the **most effective way**.
- **Common Applications**
- Logistics & production planning (e.g., best way to schedule deliveries).
- Financial planning (e.g., best investment strategy).
- Work shift planning (e.g., assigning staff to shifts).
- Marketing campaign planning (e.g., choosing the most effective ads).
- Price determination (e.g., setting product prices).

- **Types of Optimization Models**
- **Linear Optimization** – problems with straight-line relationships.
- **Integer Optimization** – decisions must be whole numbers (e.g., number of workers).
- **Convex Optimization** – problems with curved but well-behaved relationships.
- **Network Optimization** – problems involving flows (e.g., transport routes, internet traffic).
- **Multiple-Objective Optimization** – balancing more than one goal (e.g., cost vs. quality).
- **In Short:**
- **Optimization models = choosing the best option with limited resources → used in logistics, finance, staffing, marketing, and pricing.**

- **Project Management Models (Simplified)**
- **What is a project?**
- A **project** is a set of connected activities working toward a specific goal.
- Examples: building a factory, creating software, launching a new product.
- **What is needed?**
- Careful **planning and control** of activities.
- Managing **resources**: people, technology, money.
- **Role of Mathematical Models**
- Mathematical models help organize and manage projects effectively.
- **Network Models**
  - Show activities and their order (which tasks depend on others).
  - Help calculate the total project time.

- **Stochastic Models (PERT)**
  - Used when activity times are uncertain.
  - Estimate project duration using probabilities.
- **Optimization Models**
  - Help allocate limited resources (like workers or money) in the best way.
  - Balance between **costs** and **time**.
- **In Short:**
- **Projects = many connected tasks → need planning, resources, and control → models help manage time, uncertainty, and resources.**

- **Risk Analysis Models (Simplified)**
- **What they are**
- Sometimes, decisions must be made **without knowing the future for sure**.
- Risk analysis models help decision makers **choose wisely under uncertainty**.
- **Example**
- A company wants to increase production:
- **Build a new plant** → high efficiency, but very expensive.
- **Expand existing plant** → cheaper, but less efficient.
- **Subcontract work** → lowest investment, but highest operating costs.
- Which option is best depends on **future demand**:
- High demand → new plant is best.
- Low demand → subcontracting is better.
- Medium demand → expansion works well.

- **Why it’s tricky**
- The decision must be made **before knowing future demand exactly**.
- Market research can give probabilities, but not certainty.
- **How risk analysis models help**
- They use **probability (Bayesian theory)** and **utility (value of outcomes)** to guide choices.
- Widely used in:
  - Technology investments
  - New product design
  - Research & development
  - Finance and real estate
- **In Short:**
- **Risk analysis models = tools for making smart decisions when the future is uncertain.**

- **Waiting Line Models (Simplified)**
- **What they study**
- Waiting line (queue) theory looks at **crowds and delays** when services have limited resources.
- Customers arrive at random times, and service times are also uncertain.
- Because of this, some customers must **wait in line**.
- **Main Components of a Waiting Line System**
- **Population** – the group of potential customers (finite or infinite).
- **Arrival Process** – how customers arrive (random intervals).
- **Service Process** – how long it takes to serve each customer (also random).
- **Stations/Servers** – the number of service points available.
- **Waiting Rules** – the order in which customers are served (e.g., first-come-first-served, or priority).

- **Why it's useful**
- Helps design systems like banks, hospitals, call centers, or ticket counters.
- Balances two costs:
  - **Service cost** (more staff/resources → higher cost).
  - **Waiting cost** (longer queues → unhappy customers).
- The goal is to find the **optimal trade-off**: enough service to keep waiting times short, but not too expensive.
- **In Short:**
- **Waiting line models = study queues → balance service cost vs. waiting cost → design efficient systems.**

Unit II – Mathematical models for decision making

Data Mining

- Advances in technologies for information collection and storage have led to the availability of vast amounts of data across diverse domains—including business, science, medicine, and public administration.
- The activities devoted to analyzing these large datasets, typically aimed at extracting actionable knowledge to support decision-making, are described by various terms such as *data mining*, *knowledge discovery*, *pattern recognition*, and *machine learning*.
- Specifically, *data mining* refers to the systematic exploration and analysis of large datasets to uncover recurring patterns, extract relevant insights, and derive meaningful rules.
- Its importance continues to grow, both in theoretical research and in practical applications.
- This chapter will outline and characterize data mining activities in relation to their investigative goals and analytical methodologies.
- It will also examine the properties of input data and present the data mining process, highlighting its distinct phases and their interconnections.

- **Definition of Data Mining**
- Data mining is best understood as an **iterative process** dedicated to the analysis of large-scale databases, with the goal of uncovering information and knowledge that is both reliable and practically useful for decision-makers and problem solvers.
- The iterative nature of this process arises from its division into distinct phases, each of which may require feedback, refinement, and revision.
- Typically, data mining is a **collaborative effort** involving domain experts and data analysts.
- Analysts apply mathematical and inductive learning models, while domain specialists provide contextual insights to ensure relevance and accuracy.
- Experience shows that effective data mining cannot be fully automated; it demands **continuous analyst involvement** across the various stages of investigation.
- Moreover, the knowledge extracted must be validated against the data to ensure accuracy and to avoid misleading or erroneous conclusions.

- Data mining is the overall process of collecting data, analyzing it, building models, and using the results to make decisions.
- Mathematical learning theory: Behind data mining are mathematical models and methods.
- These are the tools analysts use to discover new knowledge from data.
- How does it work?
- Data mining uses inductive learning → this means learning general rules from specific examples.
- Example: If a database shows that customers who buy bread often also buy butter, we can generalize this into a rule about shopping behavior.
- The goal is to take past observations and generalize them to the larger population as accurately as possible.

- Forms of results: The patterns discovered can look like:
- Linear equations If–then–else rules
- Groups (clusters)Charts or decision trees
- Where does the data come from?
- Data is usually collected for other purposes (like billing, sales records, or phone usage logs).
- Later, this same data can be reused for mining.
- This is different from classical statistics, where data is collected specifically for analysis using planned sampling methods.
- Key point:
- Data mining is often a secondary analysis — it works with data that already exists, rather than data gathered specifically for research.

- Data mining activities can be subdivided into two major investigation streams, according to the main purpose of the analysis: *interpretation* and *prediction*.
- **Interpretation in Data Mining**
- **Purpose:** Interpretation means finding **patterns in data** and turning them into **rules or insights** that experts can easily understand.
- **Value of rules:** The rules must be **new and meaningful**—not obvious facts—so they actually add knowledge about the system being studied.
- **Example (Retail industry):**
  - A company can group (cluster) customers with loyalty cards based on their shopping habits.
  - These customer segments can reveal **new market opportunities** and help design **better marketing campaigns**.

- **Prediction in Data Mining**
- **Purpose:** Prediction is about **guessing future values or events** based on data.
- **Examples:**
  - A mobile phone company might predict which customers are likely to switch to another provider.
  - A retail store might forecast sales of a product for the next few weeks.
- **How it works:**
  - Most prediction models use **variables (features)** stored in a database.
    - Example: A customer's chance of leaving may depend on their age, contract length, and how often they call other networks.
  - Some models, called **time series models**, use only **past values of the same variable** (like past sales numbers) to predict future values.
- **Extra insight:** Sometimes, models built for prediction also help with **interpretation**.
  - Example: Classification trees not only predict outcomes but also show **which factors explain patterns** in the data.
- In short: *Prediction in data mining means using past data to estimate what will happen in the future*.

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- **Models and Methods for Data Mining**
- **1. Types of Learning Methods**
- Data mining uses different methods to find patterns and make predictions. These methods come from three main areas:
- **Computer Science (Machine Learning)**
  - Examples: *Classification trees, association rules*
  - Focus: Learn from data by trial and error (empirical approach).
  - Think of it like teaching a computer by showing it many examples.
- **Statistics (Multivariate Methods)**
  - Examples: *Regression, Bayesian classifiers*
  - Focus: Use mathematical formulas and assumptions (parametric, theory-based).
  - Think of it like solving problems with equations and probability.
- **Mathematics (Statistical Learning Theory)**
  - Focus: Strong theoretical foundation, combining probability, optimization, and statistics.
  - Think of it like building a bridge between math theory and practical data mining.

- **2. Common Steps in Building a Data Mining Model**
- No matter which method you choose, the process usually involves:
- **Choosing a model type**
  - Decide how you want to represent patterns (e.g., tree, equation, rule set).
- **Defining a metric**
  - Pick a way to measure success (accuracy, error rate, etc.).
- **Designing an algorithm**
  - Create a step-by-step procedure to build the model and improve it based on the metric.
- **3. Applications**
- Different methods are best suited for different tasks:
  - **Prediction** → Regression, decision trees
  - **Pattern recognition** → Association rules, clustering
- Some methods can be used for multiple tasks, but usually one fits best for a specific activity.

- **Student-Friendly Analogy**
- Imagine data mining like preparing for an exam:
- **Machine learning** = Practice with past question papers until you spot patterns.
- **Statistics** = Use formulas and probability to predict the most likely answers.
- **Statistical learning theory** = Study the *rules of learning itself* to understand how knowledge can be built efficiently.

- **Data Mining vs. Classical Statistics vs. OLAP**
- **1. The Core Difference**
- **Data Mining** → *Active* approach: the computer finds patterns and makes predictions on its own.
- **Classical Statistics & OLAP** → *Passive* approach: humans make guesses (hypotheses or intuitions), and the tools only confirm or reject them.
- **2. How Each Works**
- **Classical Statistics**
  - Flow: *Top-down* (start with a hypothesis → test it with data).
  - Example: "Does studying 2 hours a day improve exam scores?" → Collect sample data → Confirm or reject.
- **OLAP (Online Analytical Processing)**
  - Flow: *Top-down* (start with intuition → slice and dice data cubes to check).
  - Example: A manager thinks sales are higher in December → Use OLAP to view reports and charts → Confirm or reject.

- **Data Mining**
  - Flow: *Bottom-up* (start with raw data → let models discover patterns).
  - Example: Feed in years of sales data → The system discovers that customers who buy baby products often also buy coffee → New knowledge, not just confirmation.
- **3. Why Data Mining Matters**
- With **huge datasets**, it's hard for humans to guess meaningful hypotheses in advance.
- Data mining models actively generate insights and predictions, making them critical for success in modern analytics.
- **Student-Friendly Analogy**
- Think of it like solving a mystery:
- **Statistics** = You already suspect the butler did it, so you collect evidence to confirm.
- **OLAP** = You have a hunch about the butler, so you check different reports and angles.
- **Data Mining** = You let the detective (computer) analyze all clues and *discover* that it was actually the gardener.

- **Applications of Data Mining**
- Data mining can be used in many fields to **discover patterns, make predictions, and support decisions**. Here are the main applications:
- **1. Relational Marketing**
- **Customer segmentation** → Find groups of customers likely to respond to campaigns.
- **Retention campaigns** → Identify which customers are at risk of leaving.
- **Response prediction** → Estimate how many people will say "yes" to a campaign.
- **Buying behavior analysis** → Understand why customers buy certain products.
- **Market basket analysis** → Discover products often bought together (e.g., chips + soda).
- 👉 Analogy: Like a shopkeeper noticing that people who buy baby diapers often also buy coffee.
- **2. 🚨 Fraud Detection**
- Detect suspicious activities in:
  - **Telecom** → Fake phone usage.
  - **Insurance** → False claims.
  - **Banking** → Credit card misuse, illegal transactions.
- 👉 Analogy: Like a security guard spotting unusual behavior in a crowd.

- **3. ⚖️ Risk Evaluation**
- Predict risks in future decisions.
- Example: A bank decides whether to approve a loan based on applicant history.
- 👉 Analogy: Like checking a student's past grades before predicting if they'll pass a tough exam.
- **4. 📄 Text Mining**
- Analyze unstructured text (articles, emails, web pages).
- Applications:
  - Search engines (Google).
  - Automatic classification of documents.
  - Spam filters for emails.
- 👉 Analogy: Like sorting books in a library by topic without reading each one fully.

- **5. 🖼️ Image Recognition**
- Classify and process digital images.
- Applications:
  - Handwriting recognition.
  - Face identification.
  - Photo correction filters.
  - Surveillance cameras detecting suspicious behavior.
- 👉 Analogy: Like your phone unlocking when it recognizes your face.
- **6. 🌐 Web Mining**
- Analyze **clickstreams** (pages visited online).
- Applications:
  - Improve e-commerce sites.
  - Personalize web pages.
  - Cache popular pages.
  - Evaluate online training effectiveness.
- 👉 Analogy: Like tracking footprints in the sand to see where people walked most.

- **7. 🏥 Medical Diagnosis**
- Early detection of diseases using test results.
- Image analysis for medical scans (X-rays, MRIs).
- 👉 Analogy: Like a doctor spotting hidden signs in a patient's test results before symptoms appear.

Application	What It Does	Example
Relational Marketing	Finds customer patterns	Cross-selling products
Fraud Detection	Spots illegal activity	Credit card misuse
Risk Evaluation	Predicts decision risks	Loan approval
Text Mining	Classifies text	Spam filters
Image Recognition	Identifies images	Face unlock
Web Mining	Analyzes online behavior	E-commerce personalization
Medical Diagnosis	Detects diseases	Early cancer detection

Unit II – Mathematical models for decision making

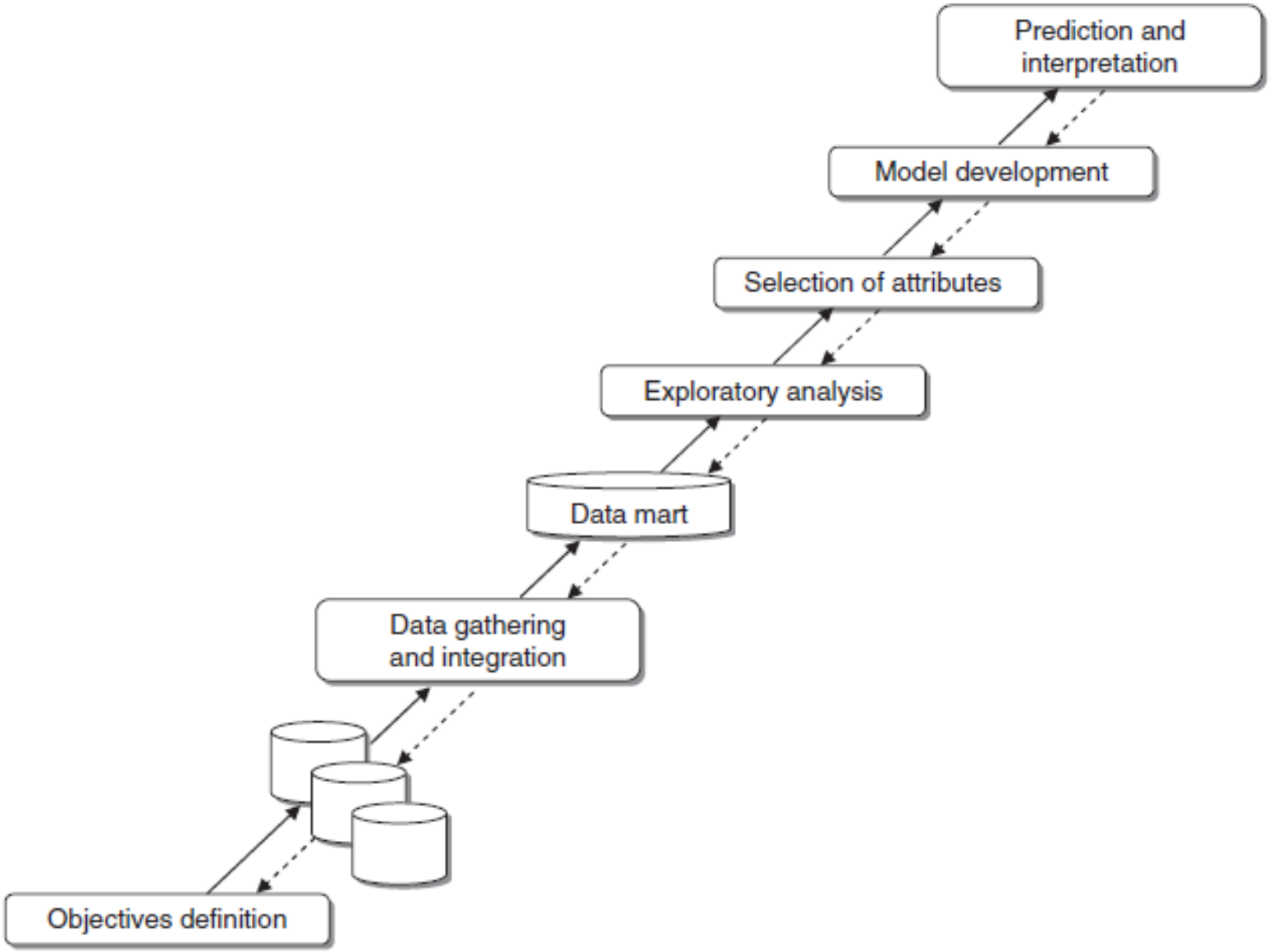
Representation of input data

- 📘 **Representation of Input Data**
- **1. What is a Dataset?**
- A **dataset** is usually a **two-dimensional table** (like a spreadsheet).
- **Rows** = observations (also called examples, cases, instances, or records).
- **Columns** = attributes (also called variables, features, or characteristics).
- 👉 Example: A dataset of students might have rows for each student and columns for age, grade, and city.
- **2. Types of Attributes**
- Attributes (columns) can be **categorical** or **numerical**:
- 🟦 **Categorical Attributes**
- Take a limited set of distinct values (usually < 100).
- Represent *qualities* (not numbers you can calculate with).
- Examples:
  - Province of residence (Mumbai, Delhi, etc.)
  - Customer loyalty (1 = left, 0 = stayed).
- ⚠️ Even if coded as numbers, you **cannot** add, subtract, or divide them.

- **Numerical Attributes**
- Take many possible values (finite or infinite).
- Represent *quantities* you can calculate with.
- Examples:
  - Monthly phone bill (₹27, ₹36, etc.).
  - You can say: “Customer A spent ₹9 less than B” or “A spent ¾ of B’s amount.”
- **3. More Detailed Categories**
- Sometimes we classify attributes more precisely:
- **Counts (Binary/Boolean)** → Yes/No, True/False, 0/1
  - Example: Does the customer have a credit card?
- **Nominal** → Categories with no order
  - Example: Province of residence (Mumbai, Delhi, Kolkata).
- **Ordinal** → Categories with a natural order but no meaningful math
  - Example: Education level (High School < Bachelor < Master).

- **Discrete Numerical** → Numbers with finite or countable values
  - Example: Number of children (0, 1, 2, …).
- **Continuous Numerical** → Numbers with infinite possible values
  - Example: Height, weight, temperature.
- **4. Mathematical Representation**
- We can represent a dataset as a **matrix**:
- m= number of rows (observations).
- n= number of columns (attributes).
- $X = [x_{ij}]$ =the whole dataset (matrix of size $m \times n$).
- **Row vector** (one record):
- $x_i = (x_{i1}, x_{i2}, \ldots, x_{in})$
- **Column vector** (one attribute across all records):
- $a_j = (x_{1j}, x_{2j}, \ldots, x_{mj})$
- 👉 Think of rows as **students** and columns as **features like age, grade, city**.

Attribute Type	Meaning	Example	Math Allowed?
Categorical	Qualitative values	Province, loyalty (yes/no)	❌
Numerical	Quantitative values	Phone bill, height	✅
Counts (Binary)	True/False	Has credit card?	❌
Nominal	Categories, no order	City names	❌
Ordinal	Ordered categories	Education level	❌ (no ratios)
Discrete	Finite numbers	Number of children	✅
Continuous	Infinite numbers	Temperature	✅



Data mining process

- 📘 **The Data Mining Process (Simplified for Students)**
- Data mining is an **iterative process** (you may go back and forth between steps). It's not just about building models — it's about preparing, testing, and applying them to solve real-world problems.
- **1. 🎯 Define Objectives**
- **What's the goal?** → Decide what problem you want to solve.
- Needs **domain experts** (people who know the business/field) + **data analysts**.
- Example: A telecom company wants to predict **customer churn** (who is likely to leave).
- **2. 📂 Data Gathering & Integration**
- Collect data from different sources (internal systems, external lists, warehouses).
- Sometimes data is already structured (easy), sometimes messy (hard).
- Risk: If data is too aggregated (e.g., only total bill per customer, not item-level purchases), you lose detail needed for analysis.
- Final format: usually a **two-dimensional table** (rows = cases, columns = attributes).
- **3. 🔍 Exploratory Analysis**
- Get familiar with the data.
- **Data cleansing**: fix errors, remove outliers, handle missing values.
- Tools: histograms (categorical data), summary statistics (numerical data).
- Example: Detect impossible birth dates or negative sales amounts.

- **4. 🧩 Attribute Selection**
- Decide which attributes (columns) are useful.
- Remove irrelevant ones.
- Create new attributes (ratios, differences, trends).
- Example: Instead of just “monthly bill,” add “change in bill compared to last month.”
- **5. 🤖 Model Development & Validation**
- Build predictive or pattern recognition models.
- Split data into:
  - **Training set** → Build the model.
  - **Test set** → Check accuracy.
- Goal: Find the best model for future predictions.
- Example: Train a decision tree on 5,000 records, test it on another 2,000.
- **6. 📈 Prediction & Interpretation**
- Use the chosen model to make predictions and generate insights.
- Integrate the model into decision-making systems.
- Example: Predict which customers are likely to churn and design retention campaigns.

- **7. 🔄 Feedback Loops**
- If results aren’t good, go back to earlier steps (e.g., redefine objectives, clean data better, select new attributes).
- Data mining is **iterative**, not linear.
- **8. 👥 Roles Involved**
- **Domain Expert** → Defines objectives, interprets results.
- **Information Systems Expert** → Manages access to data sources.
- **Statistics/ML Expert** → Builds and validates models.

Phase	What Happens	Example
Define Objectives	Set clear goals	Predict customer churn
Data Gathering	Collect & integrate data	Combine sales + customer info
Exploratory Analysis	Clean & explore data	Detect missing values
Attribute Selection	Choose/create useful features	Add “monthly bill change”
Model Development	Train & test models	Decision tree, regression
Prediction & Interpretation	Apply model to real cases	Identify at-risk customers
Feedback Loops	Iterate if needed	Redefine attributes if accuracy is low
Roles	Experts collaborate	Domain + IT + ML experts

Table 5.2 An example of input data for a data mining model

area	numin	timein	numout	Pothers	Pmob	Pland	numsms	numserv	numcall	dirept	churner
3	32	8093	45	0.14	0.75	0.12	18	1	0	0	0
3	277	157842	450	0.26	0.35	0.38	9	3	0	1	0
1	17	15023	20	0.37	0.23	0.40	1	1	0	0	0
1	46	22459	69	0.10	0.39	0.51	33	1	0	0	0
1	19	8640	9	0.00	0.00	1.00	0	0	0	0	0
2	17	7652	66	0.16	0.42	0.43	1	3	0	1	0
3	47	17768	11	0.45	0.00	0.55	0	0	0	0	0
3	19	9492	42	0.18	0.34	0.48	3	1	0	0	1
1	1	84	9	0.09	0.54	0.37	0	0	0	0	1
2	119	87605	126	0.84	0.02	0.14	12	1	0	0	0
4	24	6902	47	0.25	0.26	0.48	4	1	0	0	0
1	32	28072	43	0.28	0.66	0.06	0	1	0	0	0
3	103	112120	24	0.61	0.28	0.11	24	2	0	0	0
3	45	21921	94	0.34	0.47	0.19	45	2	0	1	0
1	8	25117	89	0.02	0.89	0.09	189	1	3	0	0
3	4	945	16	0.00	0.00	1.00	0	0	0	0	1
2	83	44263	83	0.00	0.00	0.67	0	0	0	0	1
2	22	15979	59	0.05	0.53	0.41	5	2	0	1	1
2	0	0	57	0.00	1.00	0.00	15	1	1	0	1
4	162	114108	273	0.18	0.15	0.41	2	3	0	1	1
4	21	4141	70	0.14	0.58	0.28	0	1	0	1	1
4	33	10066	45	0.12	0.21	0.67	0	0	0	0	1
4	5	965	40	0.41	0.27	0.32	64	1	0	0	1

Table 5.3 Meaning of the attributes in Table 5.2

attribute	meaning
area	residence area
numin	number of calls received in period $t-2$
timein	duration in seconds of calls received in period $t-2$
numout	number of calls placed in the period $t-2$
Pothers	percentage of calls placed to other mobile telephone companies in period $t-2$
Pmob	percentage of calls placed to the same mobile telephone company in period $t-2$
Pland	percentage of calls placed to land numbers in period $t-2$
numsms	number of messages sent in period $t-2$
numserv	number of calls placed to special services in period $t-2$
numcall	number of calls placed to the call center in period $t-2$
diropt	binary variable indicating whether the customer corresponding to the record has subscribed to a special rate plan for calls placed to selected numbers
churner	binary variable indicating whether the customer corresponding to the record has left the service in period t

- **Example 5.2 – Retention in the mobile phone industry.**
- Table 5.2 shows the two-dimensional structure of input data from an example of the analysis of customer loyalty.
- Suppose that a mobile phone company carries out a data mining analysis with both prediction and interpretation goals.
- On the one hand, the company wishes to assess the likelihood of future churning by each customer, in order to target marketing actions for retention purposes.
- On the other hand, the intent is to understand the reasons why customers churn, with the purpose of improving the service level and reducing future churning.
- Table 5.2 contains 23 observations and 12 attributes, whose meaning is indicated in Table 5.3.
- The first 11 attributes represent *explanatory* variables, while the last attribute represents the *target* variable, expressing the class of each record in relation to the objectives of the data mining analysis.
- The first explanatory variable gives personal demographic information while the rest refer to the use of the service.

- Observed values are relative to time period of index $t - 2$ for the explanatory attributes, whereas for the target variable they refer to period t .
- The difference in time placement is required in order to use the model for predictive purposes. Indeed, it is necessary to predict during the current period which customers will leave the service within 2 periods, based on the available information, in order to develop timely and effective retention actions;
- Of course, in a real application the number of available attributes is much higher, of the order of hundreds or thousands, and the number of rows representing the customers is far greater, of the order of hundreds of thousands or millions of records.
- The purpose here is to obtain an inductive model that is capable of learning from past available observations and identifying a plausible relationship between the target variable and the explanatory attributes.
- Once the model has been created based on past records, it is possible to use it to predict the target class of new records or to understand common characteristics of customers who churn compared to those who remain loyal.

Unit II – Mathematical models for decision making

Analysis methodologies

- **Analysis methodologies**
- Data mining activities can be subdivided into a few major categories, based on the tasks and the objectives of the analysis.
- Depending on the possible existence of a target variable, one can draw a first fundamental distinction between *supervised* and *unsupervised* learning processes.
- **Supervised = Prediction with a known target.**
- **Unsupervised = Pattern discovery without a target.**

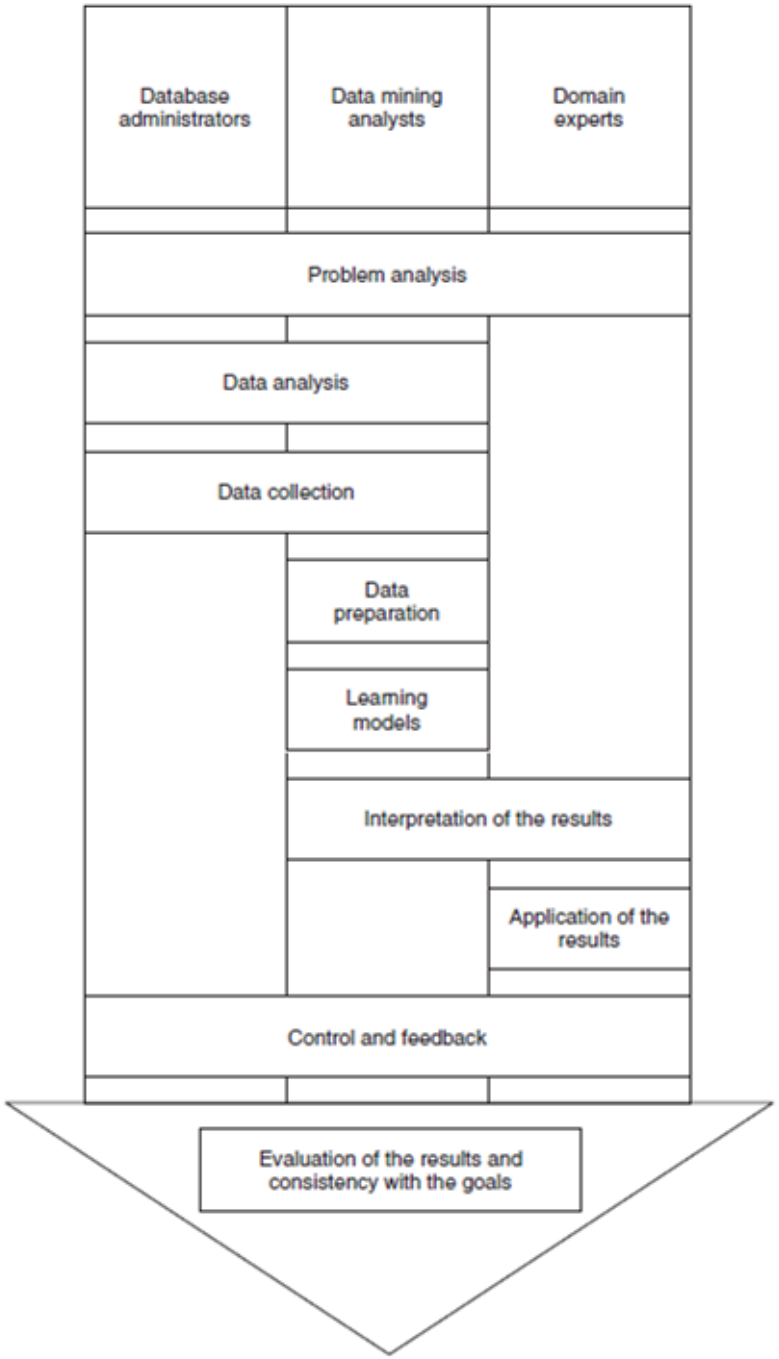
Supervised Learning

- **Definition:** Learning with a *target attribute* (something specific we want to predict).
- **Examples:**
  - Predicting if a customer is *loyal* or *not loyal* (class/category).
  - Estimating the *future value of calls* a customer will make (numeric quantity).
  - An investment firm predicting a customer's *balance sheet* using demographics and past transactions.
- **Goal:** Make predictions and interpret results based on the target attribute.

Unsupervised Learning

- **Definition:** Learning *without* a target attribute.
- **Focus:** Discovering hidden patterns, groups, or relationships in data.
- **Example:** An investment firm grouping customers into *clusters* based on similar investment behaviors.
- **Goal:** Find similarities within clusters and differences across clusters.

- Taking the distinction even further, seven basic data mining tasks can be
- identified:
- characterization and discrimination;
- classification;
- regression;
- time series analysis;
- association rules;
- clustering;
- description and visualization.



Actors and roles in a data mining process

- The first four tasks correspond to supervised data mining analyses, since a specific target variable exists that must be explained based on the available attributes or throughout its evolution over time.
- The remaining three tasks represent unsupervised analyses whose purpose is the development of models capable of expressing the interrelationship among the available attributes.
- **Characterization and Discrimination**
- **When used:** Applied when we have a **categorical target attribute** (e.g., customer type, product category).
- **Characterization**
- **Goal:** Describe what makes records in the *same class* similar.
- **Method:** Look at how attributes (like age, spending, etc.) are distributed within one class.
- **Example:** For “loyal customers,” check their average spending, age group, or call usage.

- **Discrimination**
- **Goal:** Find differences between *one class* and *other classes*.
- **Method:** Compare attribute distributions across classes.
- **Example:** Compare “loyal customers” vs. “non-loyal customers” to see what separates them.
- **How it’s done**
- Uses **exploratory data analysis** (queries, counts, charts).
- No complex learning models are needed.
- Results are shown with **histograms or charts**.
- Insights help decide which attributes are most useful for building classification models later.
- **In short:**
- **Characterization = What defines a class.**
- **Discrimination = What distinguishes one class from others.**

- **Classification**
- **Definition:** Classification is about predicting a **categorical target** (a label or class) based on known data.
- **How it works**
- You have a dataset with **observations** (records).
- Each observation has:
  - **Attributes** (features like age, customer seniority, call traffic).
  - A **target class** (e.g., active vs. churned customer).
- A classification algorithm learns from past data to **predict the class** of new observations.

- **Key points**
- The **target attribute is categorical** (finite set of values).
- Often, it’s **binary** (two classes: yes/no, active/churned).
- This is what makes classification different from **regression**, where the target is a continuous number (like predicting income or call duration).
- **Example**
- **Mobile phone company:**
  - Attributes: age, seniority, call traffic.
  - Target: whether the customer will churn (yes/no).
- The model predicts if a new customer with given attributes is likely to churn.

- **Regression**
- **Definition:** Regression is used when the **target variable is continuous** (can take many possible numeric values).
- **How it works**
- Uses explanatory attributes (features) to **predict a numeric outcome**.
- Example: Predicting **sales of a product** based on campaign type and price.
- The target can take on a wide range of values, so it's treated as continuous.
- **Relation to Classification**
- **Classification** → **categorical target** (e.g., loyal vs. not loyal).
- **Regression** → **continuous target** (e.g., probability of loyalty, sales amount).
- Sometimes, problems can be reframed:
  - A classification problem (loyal vs. not loyal) can become regression (predicting probability of loyalty).
  - A regression problem can be simplified into classification (e.g., high sales vs. low sales).

- **Time Series**
- **Definition:** A time series is when the **target variable changes over time** and values are linked to specific time periods.
- **Example:** Weekly sales of a product over 2 years → 104 data points.
- **Goal:** Use past patterns to **predict future values** (e.g., forecasting sales, stock prices, weather).
- **Key idea:** Focuses on **temporal dynamics** — how data evolves across time.
- 🔗 **Association Rules**
- **Definition:** Association rules (or affinity groupings) find **recurring relationships** between items or records in a dataset.
- **Example:** Discovering that customers often buy **bread + butter together**.
- **Uses in retail:**
  - **Shelf arrangement:** Place related products near each other.
  - **Cross-selling:** Suggest complementary items (e.g., "Customers who bought X also bought Y").
  - **Promotions:** Create bundles of products/services.
- **Goal:** Identify patterns of items that frequently occur together.

- **Clustering**
- **Definition:** A cluster is a **homogeneous subgroup** within a larger population.
- **Goal:** Divide a diverse dataset into smaller groups where members of each group are **similar to each other** but **different from other groups**.
- **Key difference from classification:**
  - **Classification** → predefined classes (e.g., loyal vs. churned customers).
  - **Clustering** → no predefined classes; groups are formed based on similarity.
- **Uses:**
  - Often a **preliminary step** in data mining.
  - Helps simplify large datasets by grouping similar records.
  - Allows further analysis to be done separately on each cluster.

- **Description and Visualization**
- **Definition:** Focuses on providing a **clear, concise summary** of information in a dataset.
- **Difference from clustering/association rules:**
  - Doesn’t group or partition records.
  - Instead, it highlights **patterns and insights** through summaries.
- **Goal:**
  - Make large datasets easier to understand.
  - Reveal hidden trends or explanations.
- **Challenge:**
  - Visualizing data meaningfully can be difficult.
  - But a **well-designed chart** makes complex information much easier to grasp.

Basic data mining tasks	Used for
Characterization and Discrimination	Categorical target attribute
Classification	Predicting categories (labels).
Regression	Predicting numbers (continuous values).
Time Series	Predicting values over time.
Association Rules	Finding items that go together.
Clustering	Grouping similar records.
Description/Visualization	Summarizing and representing data clearly.

Unit II – Mathematical models for decision making

Data preparation

- **Reliable Data for Business Intelligence**
- **Why it matters:** Business intelligence systems and decision-making models only work well if the **input data is reliable**.
- **Problem:** Data collected from primary sources and stored in a **data mart** often contains **errors or anomalies** that must be fixed.
- **Key Activities for Creating High-Quality Datasets**
- **Data Validation**
  - Detect and remove **errors, anomalies, and inconsistencies**.
  - Ensures the dataset is clean and trustworthy.
- **Data Integration & Transformation**
  - Combine data from different sources.
  - Transform it into a consistent format to improve **accuracy and efficiency** of algorithms.

- **Data Size Reduction & Discretization**
  - Reduce the number of attributes or records while keeping the dataset **informative.**
  - Makes analysis faster and easier without losing important insights.
- **Why it's important**
- These steps ensure the dataset is **high quality** and ready for business intelligence or data mining.
- Clean, integrated, and simplified data leads to **better predictions, decisions, and insights**.
- **In short:** Reliable data = better business intelligence. Validation, integration, transformation, and reduction are the core steps to achieve this.

- **Data Validation**
- Data validation ensures that input data is **clean, reliable, and consistent** before being used in business intelligence or data mining. Poor-quality data can arise from three main issues:
- **Incompleteness**
- **Problem:** Missing values in records.
- **Causes:**
  - Data not recorded systematically at the source.
  - Data unavailable during transactions.
  - Malfunctioning recording devices.
  - Data deliberately removed because it was incorrect.
  - Failure to transfer data properly from operational databases to a data mart.

- **Noise**
- **Problem:** Erroneous or anomalous values (outliers).
- **Causes:**
  - Faulty measurement, recording, or transmission devices.
  - Data expressed in different measurement units without proper conversion.
- **Impact:** Leads to inaccuracies and unreliable analysis.
- **Inconsistency**
- **Problem:** Discrepancies in data representation.
- **Causes:**
  - Changes in coding systems (e.g., product codes revised).
  - Old records not updated to match new coding schemes.
- **Impact:** Data appears inconsistent and may mislead analysis.

- **Purpose of Data Validation**
- Detect and correct **incomplete, noisy, or inconsistent data**.
- Ensure the dataset is **accurate, consistent, and ready** for analysis.
- **In short:** Data validation = cleaning missing values, removing noise, and fixing inconsistencies to make data trustworthy.

Incomplete data

- To partially correct incomplete data one may adopt several techniques.

Elimination

- **Definition:** Remove records that have missing values.
- **Key point:**
  - In **supervised learning**, if the **target attribute** is missing, the record must be discarded.
- **Limitation:**
  - If missing values are spread unevenly across attributes, eliminating records may cause a **large loss of information**.

Inspection

- **Definition:** Experts manually review missing values and suggest replacements.
- **Pros:**
  - Can be **very accurate** if done carefully.
- **Cons:**
  - **Subjective** (depends on expert judgment).
  - **Time-consuming and costly**, especially for large datasets.

Identification

- **Definition:** Assign a **special placeholder value** to mark missing data instead of deleting records.
- **Examples:**
  - For a **continuous attribute** (only positive values), use **−1** to represent missing data.
  - For a **categorical attribute**, add a new category (e.g., “Unknown”) to represent missing values.
- **Benefit:** Keeps all records in the dataset while still flagging missing values.

Substitution

- **Definition:** Replace missing values with an estimated value.
- **Methods:**
  - **Mean substitution:** Replace missing values with the average of the attribute.
    - Works only for numerical attributes.
    - Not effective if values are **skewed/asymmetric**.

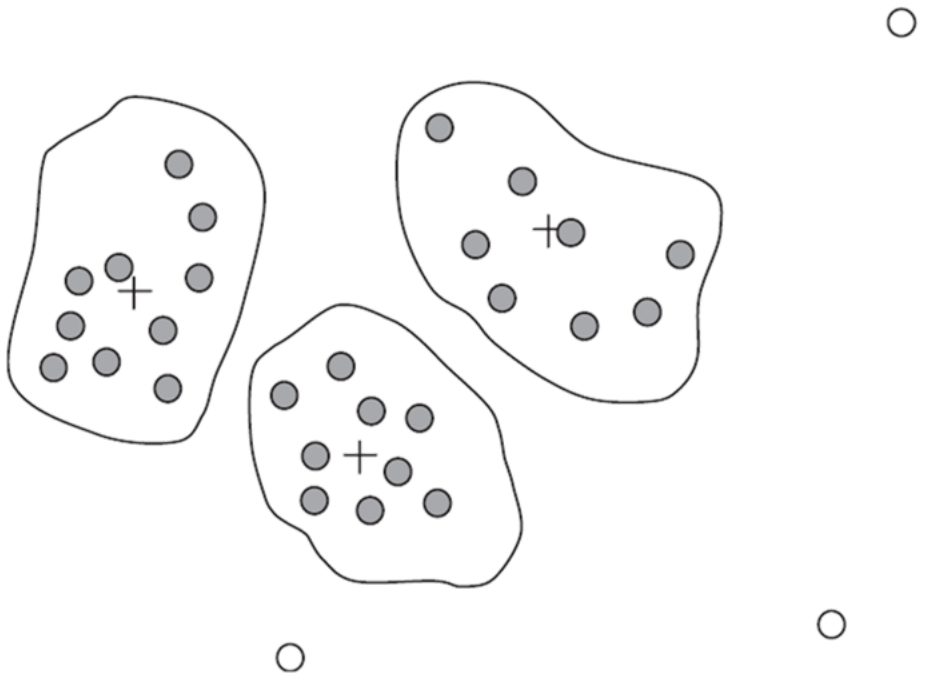
- **Methods:**
  - **Class-based mean substitution:** In supervised learning, use the mean calculated **within the same target class**.
    - Example: For churn prediction, replace missing age with the average age of customers in the same class (churned vs. active).
  - **Advanced estimation:** Use **regression models** or **Bayesian methods** to predict missing values.
    - More accurate but **complex and time-consuming**, especially with large datasets and many missing values.

In short:

- **Elimination = Quick but risky (loss of data).**
- **Inspection = Accurate but slow and subjective.**
- **Identification = Mark missing values with a special code.**
- **Substitution = Fill missing values with estimated replacements (mean, class-based mean, or advanced models).**

- **Data Affected by Noise: Outlier Detection & Regularization**
- **What is Noise?**
- **Noise** = random perturbations in numerical attributes.
- Leads to **anomalies/outliers** that distort analysis.
- Goal: **Identify** → **Correct/Remove** → **Regularize**.
- **Techniques for Identifying Outliers**
- **1. Dispersion-Based Methods**
- Compute **mean** ($\bar{\mu}_j$ **(**and **variance** ($\bar{\sigma}_j^2$**.(**
- If distribution ≈ normal:
  - Outliers = values outside:
- $\left(\bar{\mu}_j, -z_{\alpha/2}\bar{\sigma}_j\ \bar{\mu}_j + z_{\alpha/2}\bar{\sigma}_j\right)$
- Example: For $\alpha = 0.05$, ≈ 96% confidence.
- **Limitation**: Assumes bell-shaped (normal) distribution.
- **Alternative**: **Chebyshev's theorem**
- Works for *any distribution*.
- Provides broader but valid bounds.

- **2. Clustering-Based Methods**
- Group records into clusters based on **distance**.
- Outliers = points not belonging to any cluster.
- Advantage: Considers **multiple attributes simultaneously** (unlike dispersion).
- **Variant: Parametric Clustering**
- Define parameters:
  - p= percentage of dataset
  - d= distance threshold
- Observation $x_i$is an outlier if ≥ p% of records are farther than d.
- **3. Expert Judgment**
- Combine statistical detection with **domain expertise**.
- Especially important in **marketing** or applied fields.
- Prevents mislabeling rare but valid cases as noise.



Identification of outliers using cluster analysis

- **Techniques for Regularization (Correction)**
- **1. Regression-Based Correction**
- Build regression model using other variables.
- Predict plausible value for noisy attribute.
- Replace anomalies with predicted values if outside confidence interval.
- **2. Discretization & Grouping**
- Group values into **intervals/bins** based on proximity.
- Smooths anomalies by aligning them with nearby values.

Technique	Basis	Pros	Cons
Dispersion (Mean/Variance)	Normal distribution assumption	Simple, fast	Assumes bell-shaped distribution
Chebyshev's Theorem	General distribution	Works universally	Less strict bounds
Clustering	Distance between records	Multi-attribute analysis	Requires parameter tuning
Parametric Clustering (p,d)	User-defined thresholds	Flexible	Sensitive to parameter choice
Expert Judgment	Domain knowledge	Context-aware	Subjective
Regression Regularization	Prediction models	Automated correction	Needs reliable predictors
Discretization/ Grouping	Value proximity	Simple smoothing	May oversimplify data

Unit II – Mathematical models for decision making

Data transformation

- **Data Transformation in Data Mining**
- **Purpose**
- **Data transformation** = modifying raw data to improve learning model performance.
- Examples:
  - Outlier correction
  - Principal Component Analysis (PCA)
  - Standardization (normalization)
- 📏 **Standardization (Normalization)**
- Most learning models benefit from **standardization** because:
- Attributes may have different scales (e.g., income in thousands vs. age in years).
- Without scaling, large-valued attributes dominate distance-based or gradient-based models.
- Standardization ensures **fair contribution** of all attributes.

- **Popular Standardization Techniques**
- **1. Decimal Scaling**
- Formula:
- $x'_{ij} = \frac{x_{ij}}{10^h}$
- h =parameter controlling scaling intensity.
- Effect: Shifts decimal point left by $h$positions.
- Goal: Values fall within $[-, 11]$.
- **Example:** If $x_{ij} = 345$and $h = 3$, then $x'_{ij} = 0.345$.

2. Min-Max Normalization

- Formula:
- $x'_{ij} = \frac{x_{ij} - x_{\min,j}}{x_{\max,j} - x_{\min,j}} \cdot \left(x'_{\max,j} - x'_{\min,j}\right) + x'_{\min,j}$
- Where:
  - $x_{\min,j}$ =minimum value of attribute a_j
  - $x_{\max,j}$ =maximum value of attribute a_j
  - $x'_{\min,j}, x'_{\max,j}$ =desired new range (commonly [0,1] or [-1,1])
- Effect: Rescales values into a fixed range.
- **Example**: If ages range from 10–60, min-max normalization to [0,1] maps 35 → 0.5.

3. Z-Index (Z-Score Standardization)

- Formula:
- $x'_{ij} = \frac{x_{ij} - \bar{\mu}_j}{\bar{\sigma}_j}$
- Where:
  - $\bar{\mu}_j$ =mean of attribute a_j
  - $\bar{\sigma}_j$ =standard deviation of attribute a_j
- Effect: Centers data at mean = 0, scales variance = 1.
- If distribution ≈ normal, most values fall within $(-, 33)$.
- **Example**: If exam scores have mean = 70, std = 10, then score 85 → z = 1.5.

Key Takeaways

- **Standardization is essential** for fair model training.
- **Decimal scaling** is simple but less robust.
- **Min-max** is intuitive and widely used for bounded ranges.
- **Z-score** is powerful when data is roughly normal.

Method	Formula	Range	Pros	Cons
Decimal Scaling	$x/10^h$	[-1,1]	Simple, fast	Crude, ignores distribution
Min-Max	Linear rescaling	[0,1] or [-1,1]	Preserves relative distances	Sensitive to outliers
Z-Index (Z-Score)	$(x-\mu)/\sigma$	≈ (-3,3)	Handles outliers better, widely used	Assumes normality

- **Feature Extraction in Data Mining**
- **What is Feature Extraction?**
- **Definition**: A transformation process that creates **new attributes (features)** from existing ones.
- **Goal**: Capture hidden patterns, trends, or condensed information that improves model accuracy.
- **Difference from Standardization**:
  - Standardization → modifies existing values (e.g., scaling).
  - Feature extraction → generates **new variables** that enrich the dataset.
- **Examples of Feature Extraction**
- **1. Trend-Based Features**
- Dataset: Customer spending over time intervals.
- New features:
  - **Differences**: $x_{t+1} - x_t$(change in spending).
  - **Ratios**: $\frac{x_{t+1}}{x_t}$(growth rate).
- Captures **temporal trends** and customer behavior dynamics.

- **2. Complex Mathematical Transforms**
- **Fourier Transform**: Converts time-series data into frequency domain (useful for periodic patterns).
- **Wavelets**: Capture both frequency and localized time information (good for signals/images).
- **Kernel Functions**: Map data into higher-dimensional space to reveal non-linear relationships (used in Support Vector Machines).
- **3. Summarization Features**
- Combine multiple attributes into a single, informative variable.
- Example: **Image recognition**
  - Raw data = pixel values.
  - Extracted features = edge detectors, texture descriptors, or face indicators.
  - These summarize local pixel groups to make classification easier.

- Comparison: Standardization vs. Feature Extraction

Aspect	Standardization	Feature Extraction
Purpose	Scale/normalize existing values	Create new attributes
Complexity	Simple (scaling, centering)	Complex (ratios, transforms, kernels)
Output	Same number of attributes	More attributes (expanded dataset)
Example	Z-score normalization	Fourier transform, spending ratios
Benefit	Fair comparison across features	Richer representation, improved accuracy

- **Key Takeaways**
- **Feature extraction** enriches datasets by creating new, informative variables.
- It can be **simple** (differences, ratios) or **complex** (Fourier, wavelets, kernels).
- Especially powerful in domains like **time-series analysis** and **image recognition**.
- Helps models detect patterns that raw attributes alone may not reveal.

- **Data Reduction in Data Mining**
- **Why Data Reduction?**
- Large datasets → high computational cost, slower algorithms.
- **Goal**: Reduce dataset size while preserving quality of results.
- Benefits:
  - **Efficiency**: Faster computation, especially for superlinear algorithms.
  - **Accuracy**: Should not significantly compromise model performance; sometimes improves generalization.
  - **Simplicity**: Produces models that are easier to interpret and explain.

- **Criteria for Data Reduction**
- **Efficiency**
  - Smaller dataset → shorter computation time.
  - Enables testing multiple algorithms quickly.
  - Critical when algorithm complexity grows faster than linearly with dataset size.
- **Accuracy**
  - Models must remain reliable.
  - Attribute selection can improve generalization by removing irrelevant/noisy features.
- **Simplicity**
  - Important in domains where interpretability matters (e.g., healthcare, marketing).
  - Simpler rules may be preferred even at the cost of slight accuracy loss.
- ➡️ Analysts must balance **efficiency, accuracy, and simplicity** — no single technique is optimal for all cases.

Approaches to Data Reduction

- **1. Reduction in Observations (Sampling)**
- Select a representative subset of records.
- Methods:
  - Random sampling
  - Stratified sampling (preserves distribution across groups)
- Example: Instead of analyzing 10M transactions, sample 100K representative records.
- **2. Reduction in Attributes (Selection & Projection)**
- Remove irrelevant or redundant features.
- Techniques:
  - **Attribute selection**: Keep only the most informative variables.
  - **Projection**: Transform attributes into fewer dimensions (e.g., PCA).
- Example: In customer analytics, drop attributes like “middle name” that don’t affect purchasing behavior.

- **3. Reduction in Values (Discretization & Aggregation)**
- Simplify continuous or detailed values.
- Techniques:
  - **Discretization**: Convert continuous variables into categorical bins (e.g., age → {0–18, 19–35, 36–60, 61+}).
  - **Aggregation**: Summarize data (e.g., daily sales → monthly totals).
- Example: Instead of storing every second of sensor data, aggregate into hourly averages.

Direction	Technique Examples	Pros	Cons
Observations	Sampling	Faster computation	Risk of missing rare cases
Attributes	Selection, Projection (PCA)	Removes noise, improves generalization	May lose subtle info
Values	Discretization, Aggregation	Simplifies interpretation, reduces storage	May oversimplify patterns

- **Key Takeaways**
- **Data reduction** balances efficiency, accuracy, and simplicity.
- Three main directions:
  - Fewer **observations** (sampling).
  - Fewer **attributes** (selection/projection).
  - Fewer **values** (discretization/aggregation).
- Done well, it makes models faster, more interpretable, and sometimes even more accurate.

- **Sampling in Data Reduction**
- **Purpose**
- **Sampling** = selecting a subset of observations from a large dataset.
- Goal: Reduce dataset size while maintaining **statistical significance**.
- Based on **inferential reasoning**: conclusions about the whole dataset can be drawn from a representative sample.
- **Determining Sample Size**
- Must be large enough to ensure **accuracy** for learning algorithms.
- Typically, **a few thousand observations** are sufficient for most models.
- Balance between:
  - **Efficiency**: smaller dataset → faster computation.
  - **Accuracy**: sample must represent the population well.

- **Sampling Procedures**
- **1. Simple Sampling**
- Randomly select observations.
- Each record has equal probability of being chosen.
- Pros: Easy, unbiased if dataset is homogeneous.
- Cons: May not preserve proportions of critical categories.
- **2. Stratified Sampling**
- Divide dataset into **strata** (groups) based on a categorical attribute.
- Sample proportionally from each stratum.
- Ensures sample reflects original dataset’s distribution.
- Example: In customer data, preserve proportions of age groups or regions.

Multiple Independent Samples

- Create several samples of fixed size.
- Apply learning algorithms to each.
- Benefits:
  - **Robustness check**: If models remain stable across samples, conclusions are reliable.
  - **Quality assessment**: Helps evaluate sensitivity to random fluctuations.

Advantages of Sampling

- **Efficiency**: Reduces computation time (linear with sample size).
- **Accuracy**: Proper sampling maintains predictive power.
- **Robustness**: Multiple samples test stability of models.
- **Interpretability**: Easier to analyze smaller, representative datasets.

Key Takeaways

- Sampling is a **powerful data reduction technique**.
- Two main types: **simple** and **stratified**.
- A few thousand records are often enough for training.
- Multiple samples help ensure **robust conclusions**.

Feature Selection (Feature Reduction)

- **Purpose**
- **Feature selection** = eliminating irrelevant or redundant variables (columns) from a dataset.
- **Goal**: Identify the most predictive combination of attributes to explain the phenomenon under study.
- **Benefits**:
  - Faster learning (fewer columns → reduced computation).
  - More accurate models (less noise, better generalization).
  - Easier interpretation (simpler rules, clearer insights).

Categories of Feature Selection Methods

- **1. Filter Methods**
- **Independent of learning algorithm.**
- Select attributes before training, based on statistical metrics.
- Example: Correlation with target variable.
- **Pros:**
  - Fast, scalable to very large datasets.
  - Flexible (can be paired with any algorithm).
- **Cons:**
  - Ignores interactions between features and the learning algorithm.

Categories of Feature Selection Methods

- **2. Wrapper Methods**
- **Dependent on the learning algorithm.**
- Evaluate subsets of attributes by training the model and measuring accuracy.
- Search engine explores combinations of variables.
- **Pros:**
  - Tailored to the specific algorithm → higher accuracy.
- **Cons:**
  - Computationally expensive (training required for each subset).
  - Exhaustive search impractical (2^n subsets for n attributes).
  - Often uses **heuristics** (greedy search).

- **Wrapper Search Schemes:**
- **Forward (Bottom-up):**
  - Start with no attributes.
  - Add one at a time based on relevance.
  - Stop when excluded attributes fall below threshold.
- **Backward (Top-down):**
  - Start with all attributes.
  - Remove one at a time based on relevance.
  - Stop when remaining attributes exceed threshold.
- **Forward-Backward (Hybrid):**
  - At each step: add best excluded attribute, remove worst included attribute.
  - Thresholds guide stopping.

- **3. Embedded Methods**
- Feature selection occurs **within the learning algorithm itself**.
- Example: **Decision Trees**:
  - At each node, evaluate predictive value of attributes.
  - Automatically select relevant attributes for splitting.
- **Pros**:
  - Efficient, integrated into model building.
  - Often yields high accuracy.
- **Cons**:
  - Tied to specific algorithms (less general).

Method	Basis	Pros	Cons	Best Use Case
Filter	Statistical metrics (e.g., correlation)	Fast, scalable, flexible	Ignores algorithm-specific interactions	Very large datasets
Wrapper	Model accuracy with subsets	High accuracy, algorithm-specific	Computationally expensive	Moderate-sized datasets
Embedded	Built into algorithm (e.g., trees)	Efficient, automatic	Algorithm-dependent	When using tree-based or regularized models

Unit II – Mathematical models for decision making

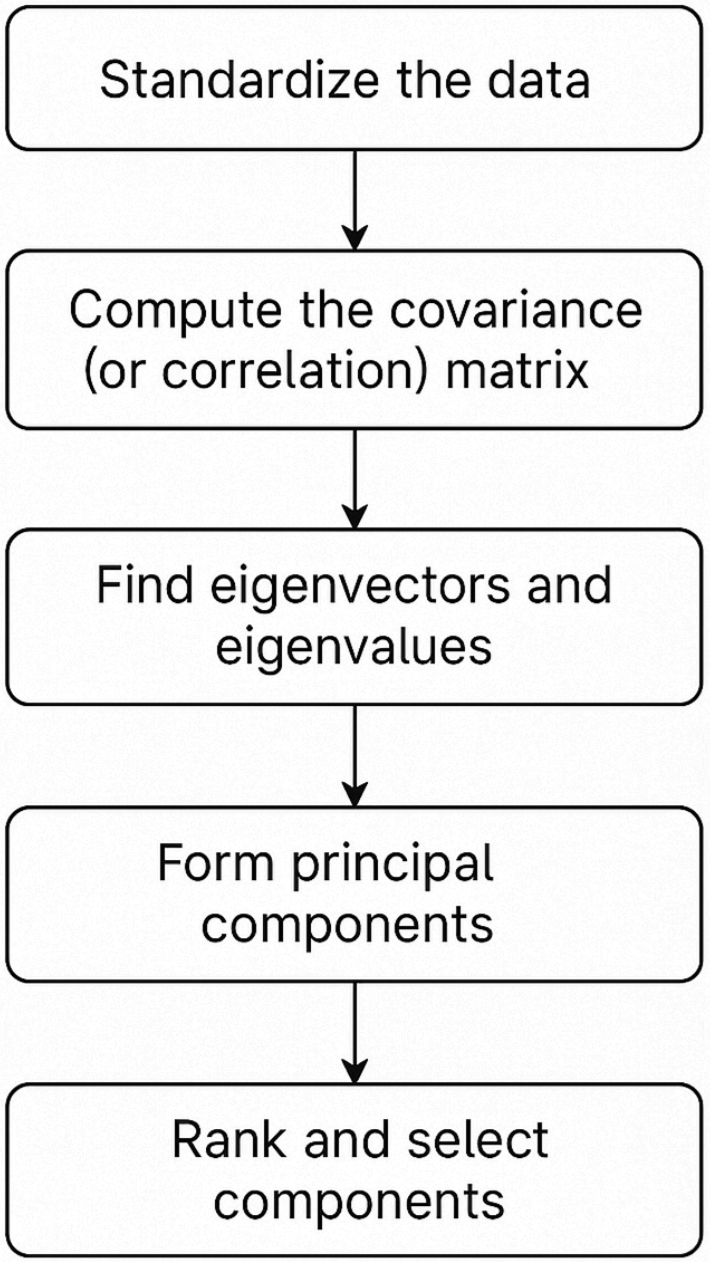
Data transformation

- **Principal Component Analysis**
- **Purpose**
- PCA reduces the number of variables (attributes) in a dataset while keeping most of the important information.
- It creates new variables (called **principal components**) that are combinations of the original ones.
- These components capture the **main patterns of variation** in the data.
- **Steps in PCA**
- **Standardize the data**
  - Make sure all variables are on the same scale (e.g., mean = 0, range similar).
  - This prevents large-scale variables from dominating.

- **Compute the covariance (or correlation) matrix**
  - This shows how variables vary together.
  - It's the foundation for finding directions of maximum variation.
- **Find eigenvectors and eigenvalues**
  - **Eigenvectors** = directions of maximum variance (principal components).
  - **Eigenvalues** = amount of variance explained by each component.
  - The first component explains the most variance, the second explains the next most, and so on.
- **Form principal components**
  - Each principal component is a linear combination of the original variables.
  - Example: $PC_1 = u_{11}a_1 + u_{12}a_2 + \cdots + u_{1n}a_n$where coefficients u_{ij}show how much each original variable contributes.
- **Rank and select components**
  - Order components by eigenvalue (variance explained).
  - Keep the first $q$components that explain most of the variance.
  - The percentage of variance explained is:
- $I_q = \frac{\lambda_1+\lambda_2+\cdots+\lambda_q}{\lambda_1+\lambda_2+\cdots+\lambda_n}$

- **Key Insights**
- **First component** = captures the strongest trend in the data.
- **Second component** = captures the next strongest trend, but orthogonal (independent) to the first.
- **Orthogonality** = components don't overlap; they explain different aspects of variation.
- **Dimensionality reduction** = instead of using all $n$variables, you can use fewer $q$components while keeping most of the information.
- 🌟 **Why PCA is useful**
- Simplifies complex datasets.
- Removes redundancy (correlated variables).
- Improves efficiency and sometimes accuracy of machine learning models.
- Helps visualize high-dimensional data in 2D or 3D plots.
- 👉 In short: **PCA finds new axes (principal components) that capture the most variation in your data, lets you drop the less important ones, and keeps the dataset simpler but still informative.**

Principal Component Analysis



- **What Is Data Discretization?**
- **Data discretization** is a technique used to reduce the number of distinct values in a dataset. It transforms:
- **Continuous data** → into **categorical classes**
- **High-cardinality categorical data** → into **simpler categories**
- **Why Use Discretization?**
- Makes models **simpler and more interpretable**
- Improves **generalization** in predictive models
- Helps generate **clearer decision rules**
- **Example:**
- Instead of:
- “If spending is in [12.21, 14.79] euros and province is B…”
- You get:
- “If spending is medium low and region is A…”
- Much easier to understand!

- **Common Discretization Techniques**
- **1. Subjective Subdivision**
- Based on expert judgment
- Example: Define spending levels as low, medium, high
- **2. Subdivision into Classes**
- **Automated grouping** of values into K classes
- Two main methods:
  - **Equal Size**: Each class has ~same number of data points
  - **Equal Width**: Each class spans equal value ranges
- **3. Hierarchical Discretization**
- Uses **concept hierarchies**
- Example: Replace “province” (100 values) with “region” (20 values)

- **How Subdivision into Classes Works**
- **Sort values** of attribute a_j
- **Choose number of classes** K
- **Group values:**
  - Equal size $\rightarrow$ distribute values evenly
  - Equal width $\rightarrow$ divide value range into equal intervals
- **Replace values:**
  - With class **average**
  - Or nearest **boundary value**
- **Benefits**
- Reduces complexity
- Enhances interpretability
- Supports better rule generation in models like decision trees

Example – Subdivision into equal size classes. Ordered values of $\mathbf{a}_j$: 3, 4, 4, 7, 12, 15, 21, 23, 27.

Class 1: 3, 4, 4

Class 2: 7, 12, 15

Class 3: 21, 23, 27

Example – Regularization by the mean value. Ordered values of $\mathbf{a}_j$: 3, 4, 4, 7, 12, 15, 21, 23, 27.

Class 1: 3.66, 3.66, 3.66

Class 2: 11.33, 11.33, 11.33

Class 3: 23.66, 23.66, 23.66

Example – Regularization by boundary values. Ordered values of $\mathbf{a}_j$: 3, 4, 4, 7, 12, 15, 21, 23, 27.
Class 1: 3, 4, 4
Class 2: 7, 15, 15
Class 3: 21, 21, 27

Example – Subdivision into equal width classes. Ordered values of $\mathbf{a}_j$: 3, 4, 4, 7, 12, 15, 21, 23, 27.
Class 1 - interval [3, 11): 3, 4, 4, 7
Class 2 - interval [11, 19): 12, 15
Class 3 - interval [19, 27]: 21, 23, 27